

CHAPTER 5

GENERAL TRAINING AND INJURY PREVENTION



PHOTO: TERJE AAMODT

Rannveig Aamodt aiming for the next hold in Lofoten, Norway.

TRAINING BASIC PHYSICAL skills such as strength, endurance and coordination provides us with a solid foundation for climbing-specific training and makes us more equipped to cope with the total training load.

Still – general training is not the same as specific training, and the extent to which general physical exercise directly affects climbing performance is unclear. There are countless examples of exceptionally skilled climbers who have been in shockingly poor physical shape, but today we see a greater focus on general training for climbers at all levels, and it is difficult to argue against this being a positive development. With younger climbers, we can incorporate more general training as a natural part of a training week. This ensures that they develop a versatile training base and coordination skills, as well as being able to withstand tougher and more climbing-specific training as they progress. Adult climbers can also benefit from doing general training in combination with regular climbing sessions, as a good training base allows for a higher training load and lowers the risk of injury.

There are many reasons why injuries occur, and being able to fully eliminate any and all risk of injury while also training to get better is utopian. Nevertheless, there are measures we can take to reduce the risk of injuries occurring, and the most important measures are strength training and how we manage the training load.

In this chapter, we look at how you can train general strength from an injury prevention perspective, how to manage your training load and we look at some of the most common climbing-related injuries.

**‘GETTING STRONG IS EASY. GETTING STRONG WITHOUT
GETTING INJURED IS HARD.’**

WOLFGANG GÜLLICH

GENERAL STRENGTH TRAINING

Strength training is an important method for reducing the risk of injury, and it is estimated that for a number of sports this risk can be reduced by as much as a third if we perform strength training on a regular basis. For example, the risk of a knee ligament injury in handball has been shown to be halved with specific strength training, and the risk of shoulder injury in handball can be reduced by a third with a specific exercise programme carried out twice a week. We can assume that this also applies to climbing-related injuries, and here both the specific strength training, which we presented at the beginning of the book, and more general strength training are important.

In the chapter on physical training, we presented several specific strength training exercises for the upper body, arms and fingers, and we explained why these exercises can help to take your climbing to the next level. By regularly performing these exercises we can assume that the risk of injury is reduced because the muscles, tendons, ligaments and bones become stronger, and we gradually build the strength required to endure the increased training load. This, in turn, prevents unnecessary breaks in our training caused by injury, breaks which might lead to large fluctuations in the training load. There is no reason not to do strength training.

General strength is especially important for training opposite movement patterns to those we encounter in climbing. In climbing, and through specific strength exercises, we train what are called the ‘prime movers’ for climbing – that is, the musculature that grips and pulls us into and up the wall. Climbing is a sport with a varied movement pattern, but this is the essence of climbing. The more we climb and train for it, the stronger the prime movers naturally become. This increases the need to strengthen the muscles that make the opposite movements – the so-called antagonists. Essentially these are muscles that push us away, rotate our shoulders outwards and pull our shoulders backwards. Important in preventing injury is creating a balance between the muscles that pull us up and into the wall, and the muscles that do the opposite.

By combining general strength training with climbing-specific exercises, you will become better equipped physically to endure an increasing training load and create a balance between the prime movers and the antagonists. Here we present a selection of the general strength exercises we believe are most important for you as a climber.

Strength training doesn’t need to be complicated or time-consuming, nor does it require fancy equipment. The most important thing is that you are aware of what exercises you should do and how to do them. We have divided the exercises into exercises which use only your own body weight (including exercises using a bungee), and more traditional strength training exercises with weights, slings and other exercise equipment.



PHOTO: MARIA DAVIES SANDBU

Slopestyle (Font 8b), Rogaland, Norway.

BODY WEIGHT EXERCISES

While we recommend that strength training using additional weights should be done on its own, body weight exercises can be performed as part of the warm-up or at the end of a climbing session. As with climbing-specific strength training, we recommend starting general strength training with a higher number of repetitions at lower loads so you can learn the techniques and movements. Then, as you progress, you can increase the load and reduce the number of repetitions per set. We have mainly chosen exercises that activate several muscle groups at the same time because this reflects the complexity of climbing, where we use multiple parts of the body. We have also chosen exercises that are primarily aimed at muscles that work in the opposite direction to what we encounter in climbing, so as to create balance in our training.





1. PUSH-UPS

Push-ups are a good exercise for your shoulders, arms and chest. For the elbows and shoulders, push-ups train the opposite movements to what we encounter in climbing – pushing away rather than pulling in – and can therefore help to reduce the risk of injury related to these joints. It's an easy exercise to start with, and it's also easy to make it harder and more challenging.

Begin by standing on all fours with your knees on the ground and your hands about shoulder width apart. Lower your chest to the floor and push back up again without twisting your elbows out. When you're able to do three sets of 10 repetitions, you can up the difficulty by standing on your toes instead of on your knees. Do as many as you can each set, and complete three sets with a 2- to 3-minute rest between each set. When you're able to do more than 10 repetitions for three sets on your toes, you can start to vary the exercise by adding rotations, as seen in photos 3 and 4.



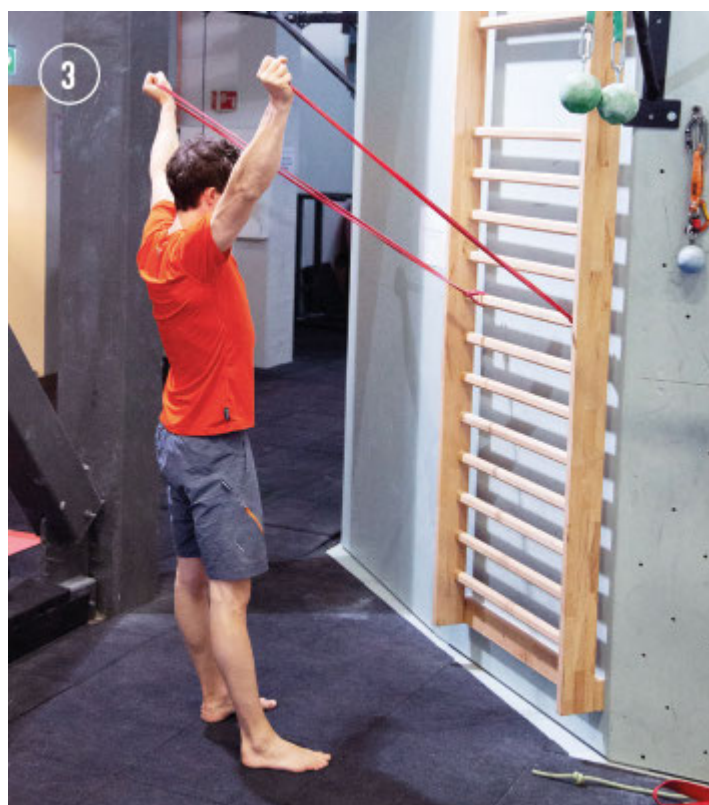
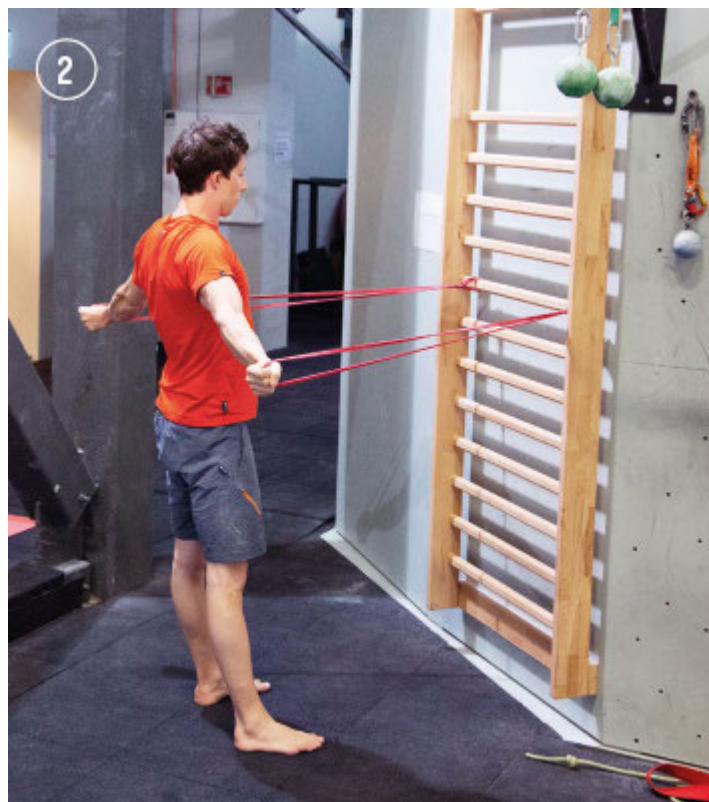


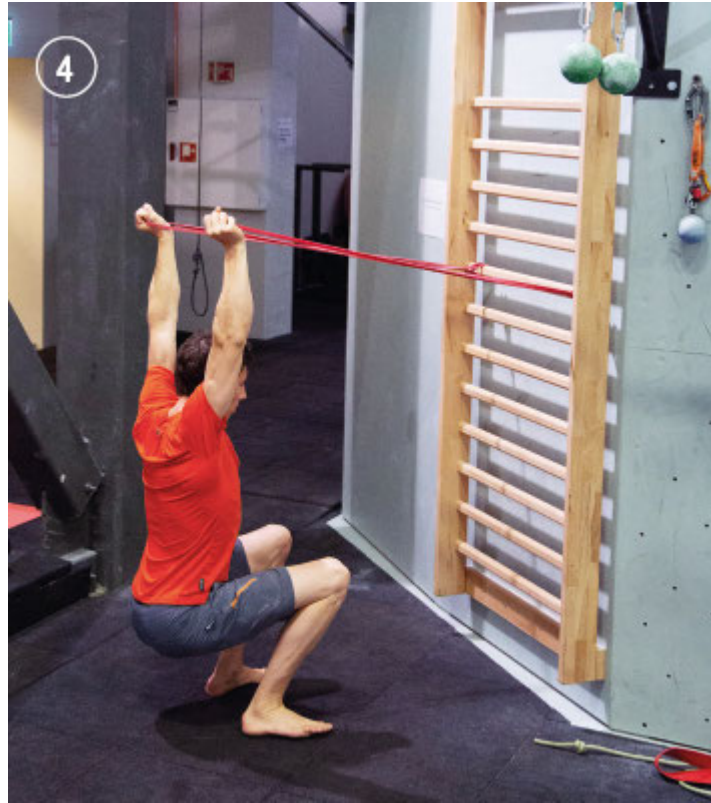
2. THE LARVA

The larva is a good exercise for your shoulders, chest, abs and back. The further you walk your hands in front of you, the more you train and utilise the full range of motion in your shoulders; it is particularly at the far end of this range that you need to build strength to reduce the risk of injury to your shoulders.

Start the exercise with both hands and feet touching the floor. Walk your hands as far in front of you as you can while keeping a strong core, then walk your feet back up to the starting position. Avoid dropping your hips or arching your back – you should feel your abdomen and back muscles working to keep your core strong, stable and straight.







3. DEEP SQUATS

Deep squats are – as the name implies – squats where you go as deep down to the ground with your hips as you can. This is a good exercise for building strength for high foot placements and it's good for training your thighs, glutes and back. When climbing your knees are constantly exposed to torsional strain as you angle your legs and feet into different positions in relation to the holds. By regularly training your knees you can build strength around the knee joints and thereby reduce the risk of injury. To make the exercise harder and at the same time incorporate training for your shoulders and upper body, we recommend using a bungee and pretending you have the wings of an angel.

Start the exercise with your arms over your head and the bungee attached in front of you. Move your arms down to the side and back up again as if you're making snow angels. When your arms are back above your head, do a deep squat while keeping your arms pointing straight up; repeat the angel movement when you're standing upright again. You should feel the muscles working actively in your lower back, thoracic spine, between your shoulder blades, in your thighs and in your glutes.





4. LUNGES WITH ARM RAISE

Lunges with arm raise is a similar exercise to the deep squats, but here you are activating the muscles in your back and shoulder differently. Raising your arm up and back with added resistance is a good way to train the opposite motion to a typical climbing move. In addition to the balance element, lunges train your glutes and the fronts of

your thighs. The exercise starts in an upright position with one leg in front of the other. Shift your weight forwards to the front leg and lower your hips. As you stand back up, raise your arm with your thumb pointing up. With your left foot forwards, lift your right hand, and vice versa. You should feel the muscles working actively in your lower back, thoracic spine, between your shoulder blades as well as in your thighs and glutes.



PHOTO: MIKE HUTTON

Anna Shepherd on The Crystal Method (Font 7b+), Caley, West Yorkshire, England.

Below is a suggestion for a strength training routine to complete after a climbing session. If you want to use some of the exercises as part of your general warm-up you can do 1–2 sets of each exercise at low loads.

PROGRAMME

EXERCISE	REPETITIONS	SETS	REST BETWEEN SETS
----------	-------------	------	-------------------

Push-ups	8–10	3–4	2–3 minutes
Larva	8–10	3–4	2–3 minutes
Deep squats	8–10	3–4	2–3 minutes
Lunges with arm raise	8–10 per side	3–4	2–3 minutes

SUSPENSION TRAINING

Training with slings has become a popular form of training over the last few years, and it's a good training method for combining our own body weight with unstable ground. The big advantages for climbers are that we can train close to the limit of the range of our joint mobility, it requires good muscular control over the different joints because the slings hang freely, and all the exercises challenge our core muscles.

As a general rule for all three exercises here, we recommend that if you can complete 10 repetitions for all three sets, you should increase the difficulty by standing on your toes instead of on your knees. Then work your way back up to three sets of 10 repetitions with 2 to 3 minutes of rest between each set. It's important to maintain control throughout the movements and to stabilise your shoulders and arms. The three exercises we recommend are:



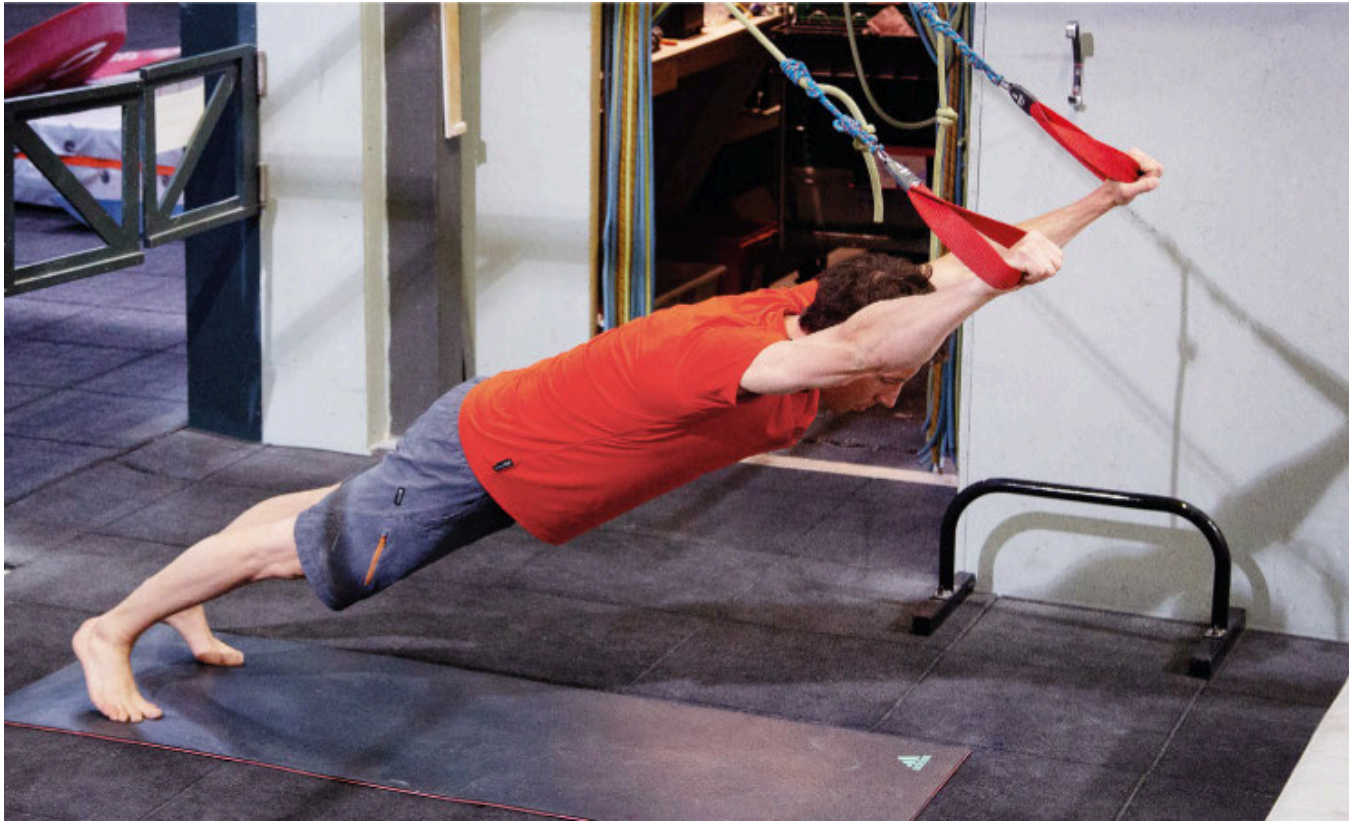
1. PUSH-UPS

Begin by standing on your knees and grab the slings with your hands at shoulder width. Lower your chest to the same height as your hands and push back up.



2. CHEST FLIES

Begin by standing on your knees and grab the slings with your hands at shoulder width. Move your arms out to each side and lower your chest. Push yourself back up by pressing your arms back together.

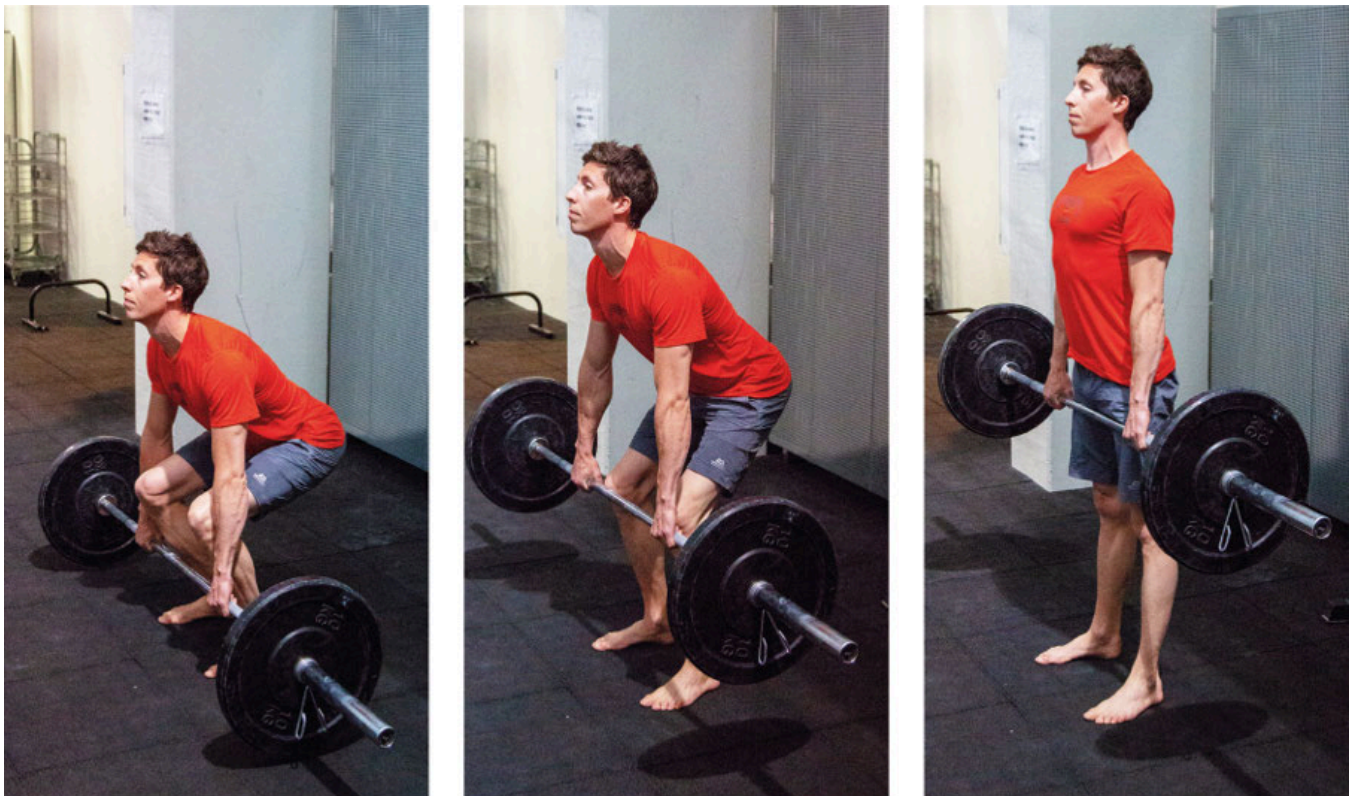


3. SUPERMAN

Begin by standing on your knees and grab the slings with your hands at shoulder width. Move your arms straight ahead while tightening your abdominal muscles and leaning forward. Stop when your arms are aligned with your ears. Push yourself back up by pressing your arms down.

WEIGHT TRAINING

There are many good exercises which rely on your body weight alone, but we still recommend training with additional weights as a supplement to your climbing. Training with weights is an effective way to build muscle strength and will have a positive effect on both your general physique and risk of injury. The only downside of weight training is that muscle growth increases body weight, which isn't always beneficial in a sport like climbing. But still, this is a relatively small problem for most of us, and as long as weight training remains a supplement to, and not a substitute for, climbing, we see it as a positive way of building a solid base physique.



Remember to tighten your abdomen and keep your back straight throughout the exercise.

1. DEAD LIFT

The dead lift is one of the best exercises for strengthening your lower back and is a basic strength exercise for most sports. Some of you might have been told that it's harmful to your back; this myth has been debunked, though it is important to start easy with lower loads and a higher number of reps so you can properly learn the technique.

Start the exercise standing up with your arms to the sides and your feet shoulder width apart. Lift the bar straight up until you're standing upright, then lower it back down to the floor. You should feel your thighs, glutes and lower back working – and obviously your grip – when lifting heavy weights. When lifting heavy weights you can

use wrist straps or an opposing grip position (one overhand and one underhand), to stop the bar rolling out of your hands.

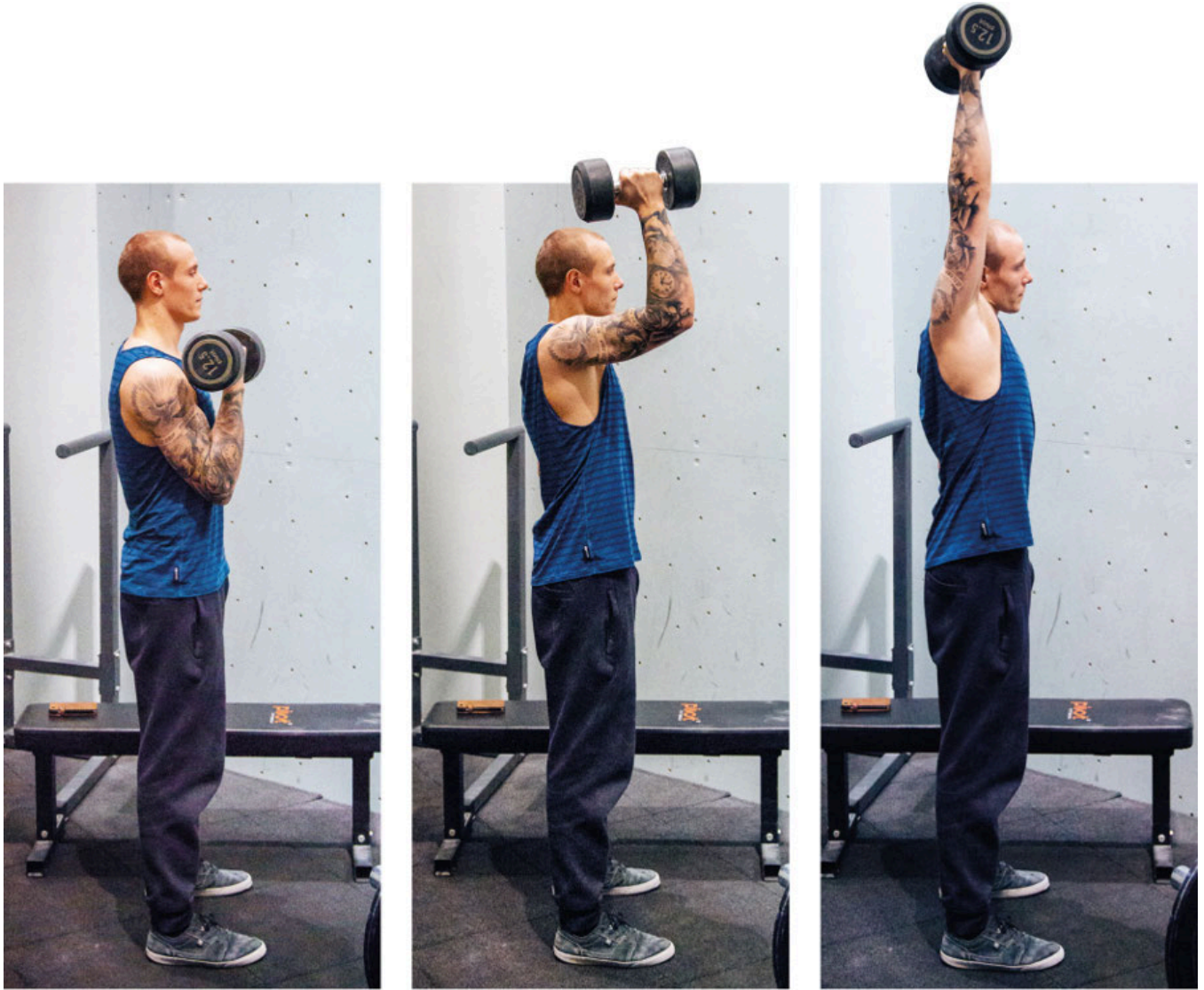


2. OVERHEAD SQUAT

The overhead squat is a good strength exercise for your shoulders, back, glutes and thighs. In addition, it requires good mobility in your shoulders and thoracic spine to keep your arms straight when you squat.

Start the exercise standing straight with the bar on straight arms above your head. Squat as deep as you can, while keeping your back and your arms straight, and stand back up. You will feel your thighs and glutes working, but mostly you will feel it in your lower back, thoracic spine and on the backs of your shoulders.

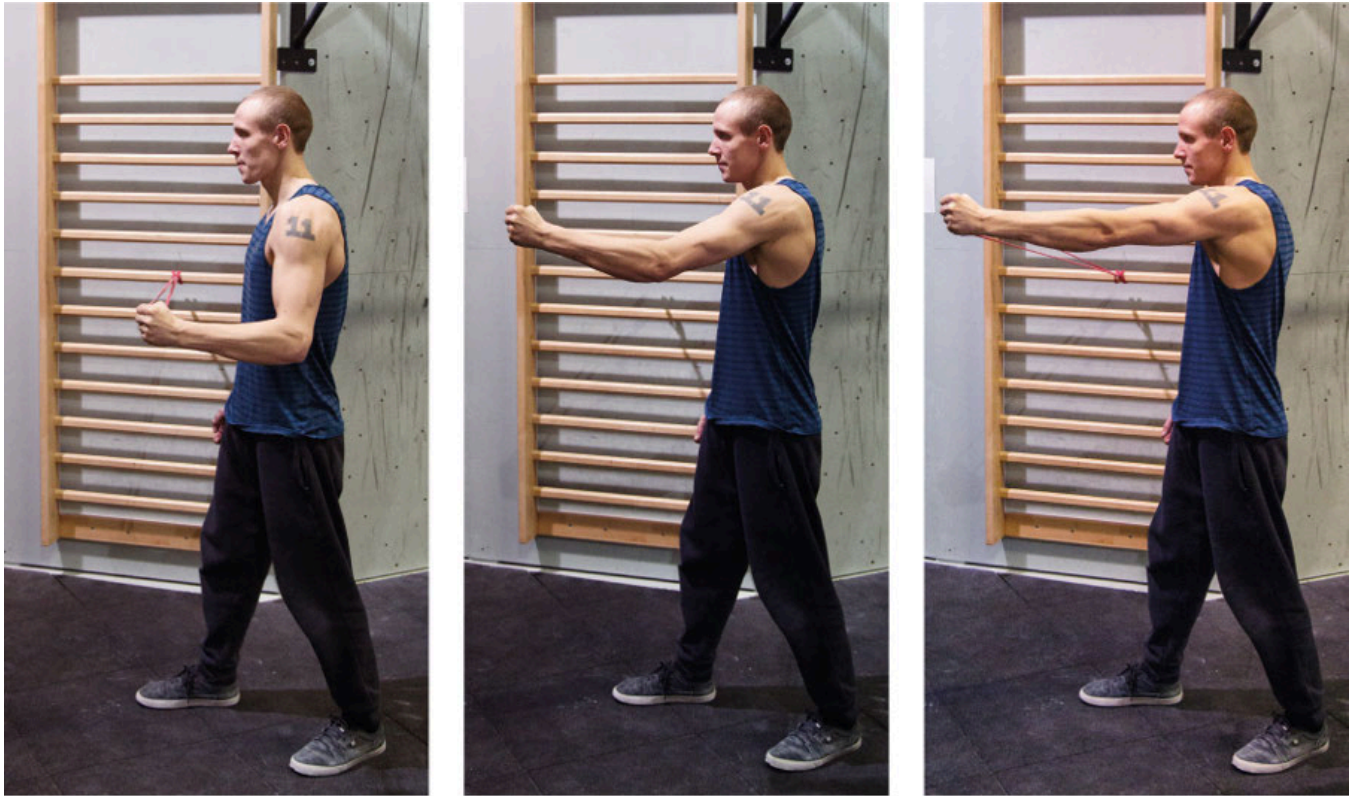
This is a difficult exercise to master, so start easy and learn the technique before adding weight. Remember to tighten your abdomen and keep your back straight throughout the exercise.



3. SHOULDER PRESS

There are many variations of the shoulder press, but the point is to add a load to the arm as we push it straight up. Here we have chosen the variation known as the Arnold press – named after its legendary inventor, Arnold Schwarzenegger – as it adds the element of shoulder rotation to the traditional shoulder press. The exercise also requires mobility in the shoulders and thoracic spine, and by correctly executing the shoulder press we can improve our mobility as well as increase our strength. To challenge the stability in your back and abdomen we recommend doing the exercise standing up, one arm at a time.

Start the exercise with your arm bent, palm facing in to your chest. Push your arm up straight above your head while rotating it so that your palm ends up facing forwards. Return to the starting position.



4. PALLOF PRESS

The Pallof press works the outside and back of the shoulder and between the shoulder blades, and it's a good way to build strength and stability around the shoulder. The exercise can be done using a bungee, but it is easier to control the progression of your training when using a weighted cable pulley system you should find at your local gym.

When training your left arm, stand with your right foot forwards, and vice versa. Start with your arm down by your side, with your forearm pointing forwards – i.e. bent 90 degrees at the elbow. Push your arm forwards, while working actively to stop it from rotating in, until it's pointing straight ahead. You should feel the muscles working on the outside and back of your shoulder and between the shoulder blades.

PROGRAMME

EXERCISE	REPETITIONS	SETS	REST BETWEEN SETS
Dead lift	8–12	3–4	More than 3 minutes
Overhead squat	8–12	3–4	More than 3 minutes
Shoulder press	8–12	3–4	More than 3 minutes
Pallof press	8–12	3–4	More than 3 minutes

This programme is a good starting point when you're just getting into weight training. By starting with a relatively high number of repetitions and sets you'll learn and practise the techniques and build a solid foundation before moving on to heavier weights. You should feel tired after every set. Adjust the number of kilos per repetition to suit the number of repetitions to ensure you are training at the right loading for your level.

If you are regularly training with weights one or more times per week, you should gradually increase the load per repetition and reduce the number of repetitions accordingly.

An example might then look like:

EXERCISE	REPETITIONS	SETS	REST BETWEEN SETS
Dead lift	3–5	3–5	More than 3 minutes
Overhead squat	3–5	3–5	More than 3 minutes
Shoulder press	3–5	3–5	More than 3 minutes
Pallof press	3–5	3–5	More than 3 minutes

We recommend keeping weight training separate from climbing sessions and preferably you should take a rest day before climbing. Strenuous strength training can come at the expense of the quality of your climbing training, and your climbing training should always take priority. If you are lifting weights several times a week, a good rule of thumb is to allow each muscle group to rest for at least two days.

Exercising using heavier weights and fewer repetitions will to a greater extent stimulate muscle activation rather than muscle growth. Therefore, after learning the exercises and training with a higher number of repetitions during the initial time period, it is advantageous for us climbers to train using heavier weights and fewer repetitions.

**‘TRAINING HARD ISN’T DANGEROUS. SPIKES IN TRAINING
LOAD ARE.’**

TIM GABBETT

LOAD MANAGEMENT

Large fluctuations in load – when periods of hard training follow immediately after periods of less exercise and at lower loads – increase the risk of injury. Continuity in your training means that you will gradually build a tolerance to more and harder training over time. It is the relationship between how hard you train and what you are trained to endure that is important. An elite climber balances on this knife-edge throughout much of the year, and it is impossible to get to the stress level these athletes are at without having trained steadily and progressively over a long period of time. Without building a good foundation, they would not have reached the level they are at. This is just as important to remember for those of us climbing at lower levels.

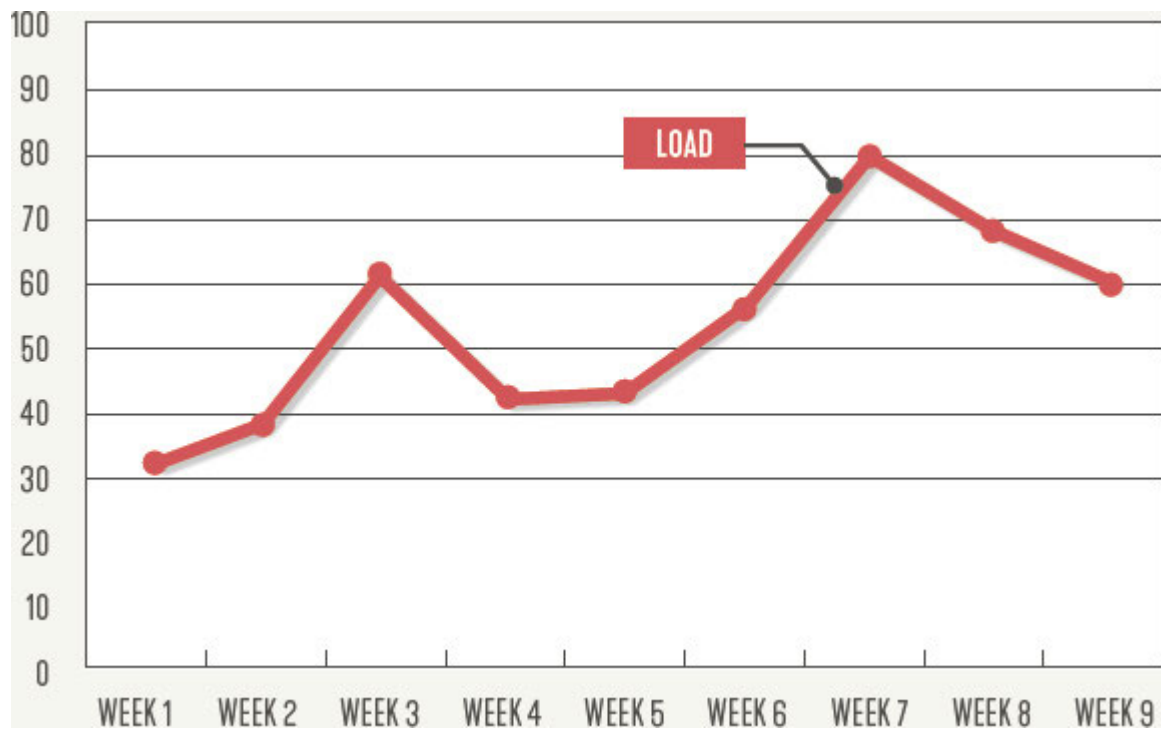
Going on a three-week holiday where you do not train or load your fingers will weaken your fingers' ability to withstand loads, and it is then important that when you get back to training you start gradually in order to reduce the fluctuations in load. If you wish to train hard after taking a long break, it's important that you set aside some time during your vacation to do a few sessions where you load your fingers and the rest of your body. A portable fingerboard is a fantastic tool in this regard. A couple of sessions of deadhangs, pull-ups, and core workouts per week during times when you're not climbing will leave you in much better shape as you go into a new training period.

There are several ways to manage load, and the easiest way is to look at the wall angles, climbing style and holds you use. Steep walls, bouldering and small holds increase the intensity for your upper body, arms and fingers significantly compared to vertical walls, lead climbing and larger holds. By alternating between bouldering and lead, physically and technically challenging climbing, as well as how steep the wall is, you can change which part of the body is doing most of the work and hence get enough rest even when training often and hard. Therefore, all that is required is an awareness of the type of climbing you are doing, and a simple training log, to get the most important injury prevention measure – load management – to work properly.

When managing load, it is useful to have some parameters to illustrate how hard you have been training. Exercise load is the product of volume and intensity. The volume is the amount of training you have done: this could be the number of minutes of training or the number of routes or boulder problems completed per session. The intensity is how hard you have been training and can be graded on a subjective scale of 0 to 10, where 10 is the highest intensity. In Chapter 2 we described intensity zones for endurance training where 5 was the highest intensity, and an endurance session in intensity zone 5 would then be a 10 on a 0 to 10 training intensity scale. Similarly, a maximum bouldering session, deadhangs and campus training would be close to a 10 on this scale, and the calculation may then look like this:

Training time (= 2 hours) x intensity (= 8) = 16.

The number 16 is an indicator of how high the training load was that session. If you do a calculation for each session in this way, you can get a number for the training load for the whole week. Large deviations from this sum over the next few weeks, either up or down, increase the risk of injury; knowing this, you can adjust the load up or down to flatten the curve and reduce the risk of injury. When you extract these values, the curve might look something like this:



Example showing training load for a competitive athlete during a nine-week base training period.

There is one obvious challenge in counting the number of training hours, and that is that usually you will not climb continuously throughout a session. With both lead and bouldering sessions, you will have shorter or longer rests, so will not distinguish between working time and resting time by counting the hours per session. You will still get a picture of how much and how hard you have been training over the course of a week, and this is the easiest way to calculate your training load.

We would like to emphasise that this is by no means an exact science and that there are many sources of error in calculating climbing training load in this way. However, in our experience, quantifying the load for a session, a week and a period is a good tool for load management. Using the examples above, you can adapt the method to your own training so that you can easily adjust the load up or down to avoid excessive fluctuations.

WHEN THE DAMAGE IS DONE

Unfortunately, when we train to improve, there will always be a risk of injury. Despite the strength training and load management, there are several factors that can lead to injury, for example, incorrect execution of a technical move. In other cases, an acute injury may simply be due to bad luck. We may experience both acute injuries, such as a sprained ankle or a torn ligament in a finger, or chronic injuries, where we do not necessarily know why they occurred or when we first noticed the symptoms.

ACUTE INJURIES

In the case of acute injuries, it is important to seek help to diagnose what has happened and the extent of the injury so that the rehabilitation can be as good as possible. It is also crucial to find out what caused the injury in the first place. Were the crash pads poorly positioned, causing you to sprain your ankle upon landing? Did your finger go 'pop' as you desperately bore down on that crimp at the end of a two-hour-long climbing session? Was it cold outside? Did you warm up? Were you tired after too much training? Tired after not getting enough sleep? There are a number of factors that can contribute to acute injuries, and by getting a sense of why the injury may have occurred, you can reduce the risk of it happening again.

For several years, the acronym PRICE has dominated decision making when acute injuries occur. PRICE is an abbreviation of Protection, Rest, Ice, Compression, Elevation, and overall, these measures should dampen the inflammatory response that follows tissue damage, thus alleviating pain and reducing swelling. However, in recent years we have seen that Rest is rarely a good strategy and that early loading of damaged tissue is crucial to shortening the time it takes to heal – and thus how long it takes before you are back in the game. Instead, we now ask the question:

‘SHOULD WE CALL THE POLICE?’

In the POLICE acronym, Rest is replaced with Optimal Loading. It's not always easy to know what optimal loading is, but it does at least point out that tissues need loading in order to heal properly. Most types of tissue in the body heal with scar tissue, and in the same way that you would want to stretch a skin wound so that the scar tissue doesn't become too short, you should do the same with muscles, tendons and ligaments. There's a big difference between complete rest versus relieving enough load so that we can utilise the injured body part without aggravating the pain and/or increasing the swelling. With a sprained ankle, we should start moving around and loading the ankle by standing and walking on it after just a few days. Although the load is reduced so that the ankle can rest, it also needs to be loaded slightly. The same goes for a finger pulley injury, where we can start with a squeeze ball, some light climbing and deadhang training relatively quickly.

CHRONIC INJURIES

Chronic injuries are different from acute injuries since we do not always know when the injury occurred or what triggered it. In most cases, a chronic injury is the result of us training:

‘TOO MUCH, TOO OFTEN, TOO FAST AND WITH TOO LITTLE REST.’

If we assume that we have been training too much, too often and with not enough rest for the tissue to adapt, it is clear that the general solution to such injuries is to better manage the load. This is not the same as taking complete rest from climbing or training, but rather looking at ways we can lower the load or load differently for a period of time. An elbow that hurts from steep bouldering can be fine with some vertical lead climbing. A finger that hurts when crimping and campus training can be fine with bouldering on large holds and open grip positions. There is a lot you can still do during periods of injury, and any training you do will help maintain both your physical shape and your technique until you are ready to train normally again. As mentioned earlier, it is also important to load the body and the injured body part to steadily build up the load capacity in the tissue after an injury. If the pain persists for a long time despite efforts to relieve and adjust the load, we recommend that you seek qualified help to clarify the cause of your discomfort. Rehabilitation from an injury and returning to the level you want to climb at can be a lot easier if you get help with finding the cause and laying down a good plan for your recovery.

By figuring out what might have caused your injury, you can adjust the load for specific areas. For example, if you have been bouldering and campusing and you’re starting to feel some pain in one finger, you can steer away from these training methods for a while, and instead focus more on technical bouldering and lead climbing, which are less stressful for your fingers. Then you can complete some shorter bouldering sessions and see how your finger responds. When you are comfortable with these sessions, you can gradually increase the load on the fingers for each session before trying some campusing again. So you calm the situation down, before gradually building it back up again. Rehabilitating an injury without correcting for the load that caused it is impossible.

Also consider that by training while injured you’re lessening the fluctuations in training load, which reduces the risk of further injury. Let’s look at an example:

Martin has injured his finger. Since he cannot train as usual, he rests for two weeks. His finger feels better after this rest, and he returns to training. His plan to gradually build his strength back up quickly disappears during his first session, and after two sessions his finger is painful again. He rests for another two weeks before the story repeats itself.

Such a response to the injury creates large fluctuations in how much the finger is loaded. Instead, Martin should have continued with easy climbing during his two weeks of rest, and then gradually increased the load. By flattening out the fluctuations, you can still train while injured, and thereby rehabilitate faster and reduce the risk of relapsing into injury.

A simple rule to remember for the treatment of chronic injury is this quote by Canadian chiropractor Greg Lehman: Calm shit down, build shit back up.



PHOTO: MIKE HUTTON

Mike Hutton traversing Hamper's Hang (Font 7a+), Stanage Edge, Peak District, England.

THE CHALLENGE OF BEING GOOD

When we have established ourselves at a certain level of climbing, we have acquired technical abilities that keep us relatively stable at this level, even after breaks in our training. Ambitious as we are, it can be a challenge to return to training at a lower level than where we left off: we are skilled enough to climb at a level that is higher than our physical condition allows. Even after relatively short breaks of just a couple of weeks, we will be significantly weakened physically, and this creates an imbalance with how much load we can withstand and what

level we are able to climb at. Many climbers start their training – whether it's following an injury or an holiday, or for other reasons – at too high a level, and thereby load too much, too often and too fast. A healthy dose of self-discipline is required to be able to start slow and gradually build back up to speed, but this will most likely reduce the number of setbacks and lead to better progression in the training.

WHAT IS PAIN?

This might seem like a strange question. We have all experienced pain at some point, but the answer might be more complicated than you think. Especially if the pain lingers for longer than expected.

You can think of pain as being the siren from an alarm system which signals danger, similar to an alarm you might have in your house. Connected to most of the tissue structures in your body are nerve fibres, whose job it is to send information to your brain about the status of the tissue. Let's call them guardians, alerting the emergency response centre – your brain – if they discover something they think is dangerous. Information regarding load, or changes in temperature, pressure or length, is information that is continuously transmitted to the response centre while we are moving. This information is interpreted and evaluated before appropriate action is initiated. Our emergency response centre is programmed to take very good care of us, and if the information it receives is considered to be dangerous or harmful to us, pain is an appropriate response to avoid further harm.

Outdoors we often encounter holds that are uncomfortable to grip. The pressure from the crystals into our fingertips makes the guardians send information to the response centre, which in turn has to decide if it is dangerous for us to keep holding on. In many cases it isn't. And, after all, we want to hang on and do the move. The best option then is to disregard the information about the crystals against our fingertips and keep on gripping. The pain response is therefore dependent on how we interpret the available information, and this interpretation is based on the following three questions:

1. Am I safe?

2. Do I have enough information to predict the future?

3. Do I have enough information to influence the future?

If the answer to all three questions is YES, then pain is an unnecessary response. Answering NO to one or more of the questions increases the need for protection, and hence also the need for pain as a response to ensure this. Let's look at two examples:

EXAMPLE 1: The day after a hard strength training session your sore muscles are sending signals via your nervous system to your brain that you are picking up a bag from the floor.

'Am I safe?' YES. I was sore just last week, and I was fine.

'Do I have enough information to predict the future?' YES. All previous experience tells me my muscle soreness will go away after a few days, and I can get back to training as normal.

‘Do I have enough information to influence the future?’ YES. I know that if I warm up well before the next session, the soreness will go away.

The conclusion is that this is not dangerous, and there is less of a need for pain as a protective response. You might still experience muscle soreness, but you probably won’t give it much thought.

EXAMPLE 2: Your fingers are aching after several hard bouldering sessions on small holds. They are sore to the touch, and gripping small holds is uncomfortable.

‘Am I safe?’ YES, probably. My fingers are painful, but I’m alive after all. I am however uncertain whether I should keep on climbing.

‘Do I have enough information to predict the future?’ NO. I don’t know if this is a serious injury and whether or not I should take a longer rest from climbing.

‘Do I have enough information to influence the future?’ NO. Since I don’t know what’s wrong, I also don’t know what I should do to get better.

When you are faced with a potentially dangerous situation which might have consequences in the form of a break from climbing – and perhaps a reduced quality of life – pain is an appropriate response until you know more about the situation. Pain is based both on signals from your body and how you interpret those signals based on your knowledge, experiences and memories, in which context this has happened, and what the consequences might entail. It is important to understand all of this so that you can understand the difference between pain and injury.

‘THE REST OF MY TEAM ALMOST DIDN’T BELIEVE ME. I INSTANTLY HEARD AND FELT WHAT IT WAS. AS I LEANED DOWN TO FEEL, AND SAID THAT I HAD TORN MY ACHILLES, THEY DIDN’T BELIEVE ME. IT WASN’T THAT PAINFUL, SO I WAS A BIT SURPRISED TO LEARN THAT IT HAD TORN COMPLETELY.’

AKSEL LUND SVINDAL, WORLD CHAMPION AND OLYMPIC GOLD MEDAL-WINNING SKIER

PAIN AND INJURY

Can injury exist without pain? What about pain without injury?

When Norwegian ski racer Aksel Lund Svindal tore his Achilles tendon during practice, a serious and potentially career-ending tissue injury, it wasn't particularly painful. We can question why this was the case, but it is just one of many examples of it being possible to sustain a severe injury without the accompaniment of any significant pain.

The opposite can also be the case – we can experience pain without tissue damage. A classic example of this is the story of the carpenter who stepped on a nail:

A carpenter stepped on a nail while at work. The nail went right through his shoe and into his foot. He was, naturally, in great pain, and was rushed to the hospital for further examination. With his foot still in the shoe, they took X-ray photos, but it turned out the nail had passed right between the carpenter's toes and hadn't actually damaged any part of his foot.

So, tissue damage isn't a necessary prerequisite for experiencing pain. All it takes is the potential of an injury and a situation that is perceived as dangerous. Pain is a protection mechanism and probably one of the most important survival mechanisms we have, and it is obviously vulnerable to misinterpretation.



